



BLOCK URI-205

Lecture (I)



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Learning objectives

After this lecture, students should be able to:

Describe the histological structure of the kidney; cortical and medullary organization . .

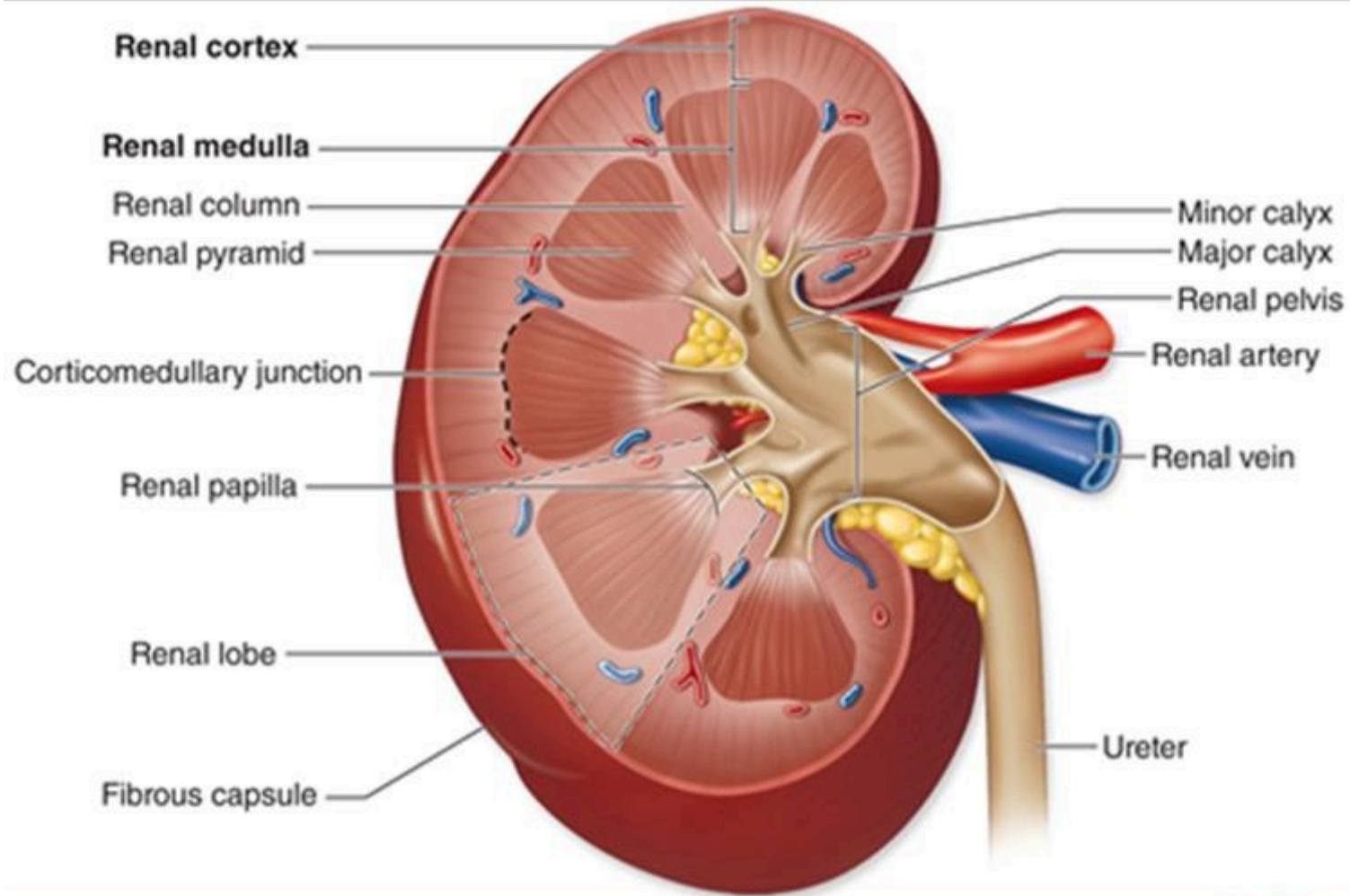
- Discriminate the different parts of functional units; the nephrons . .

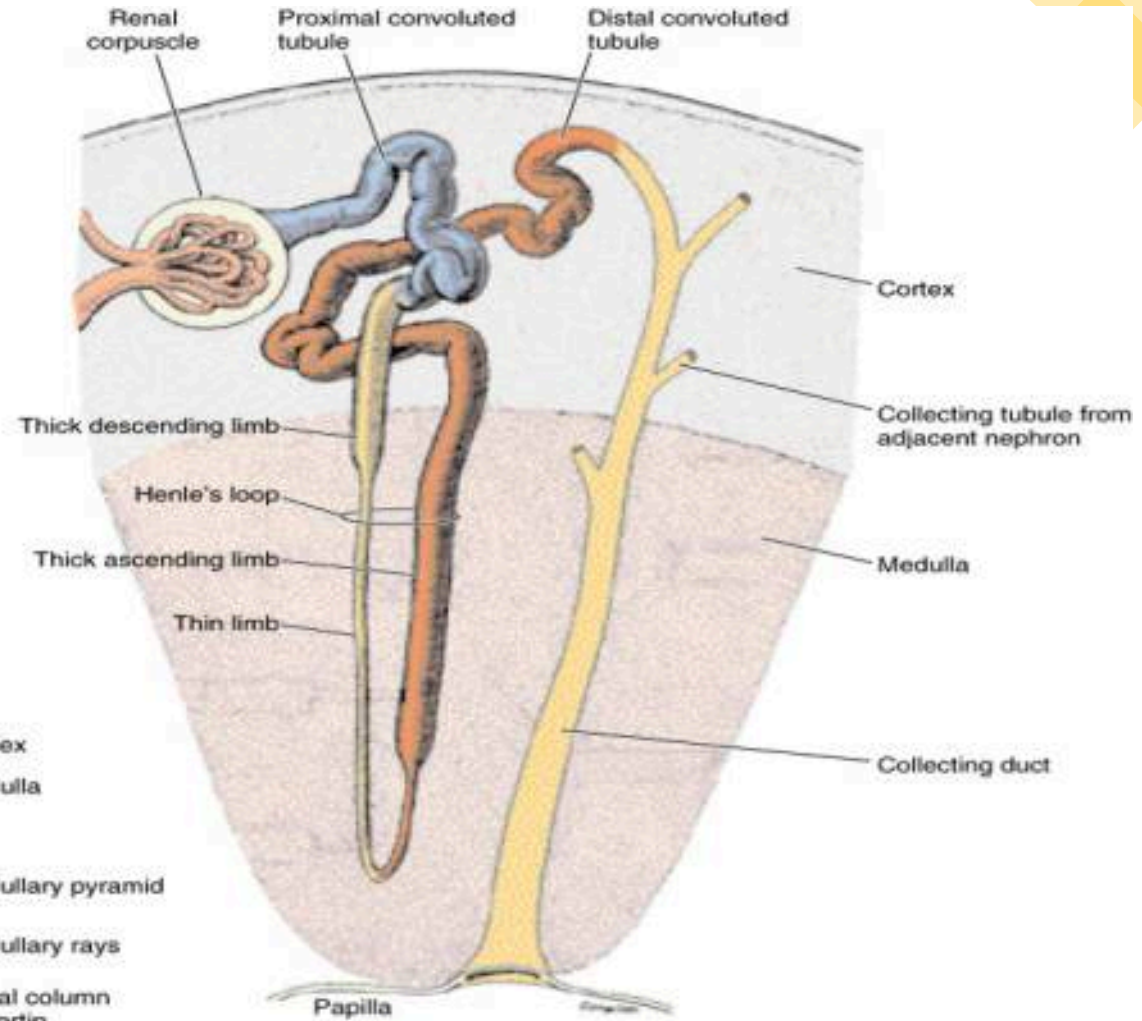
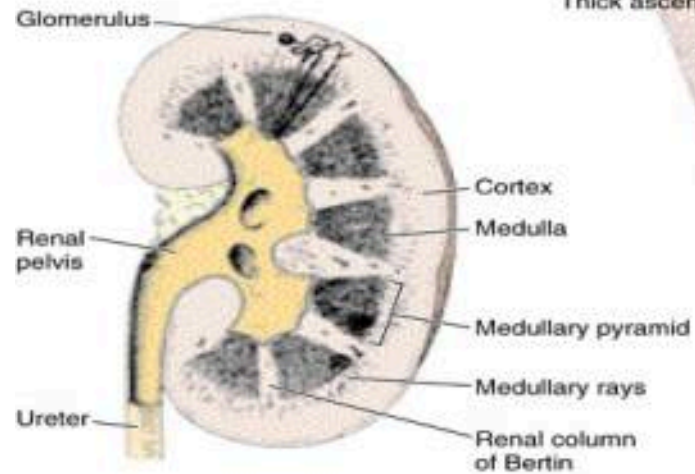
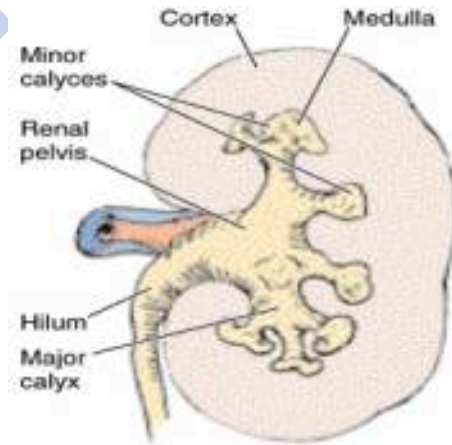
- Describe the histological structure of the renal corpuscles and correlate the histological structure and functions of these corpuscles

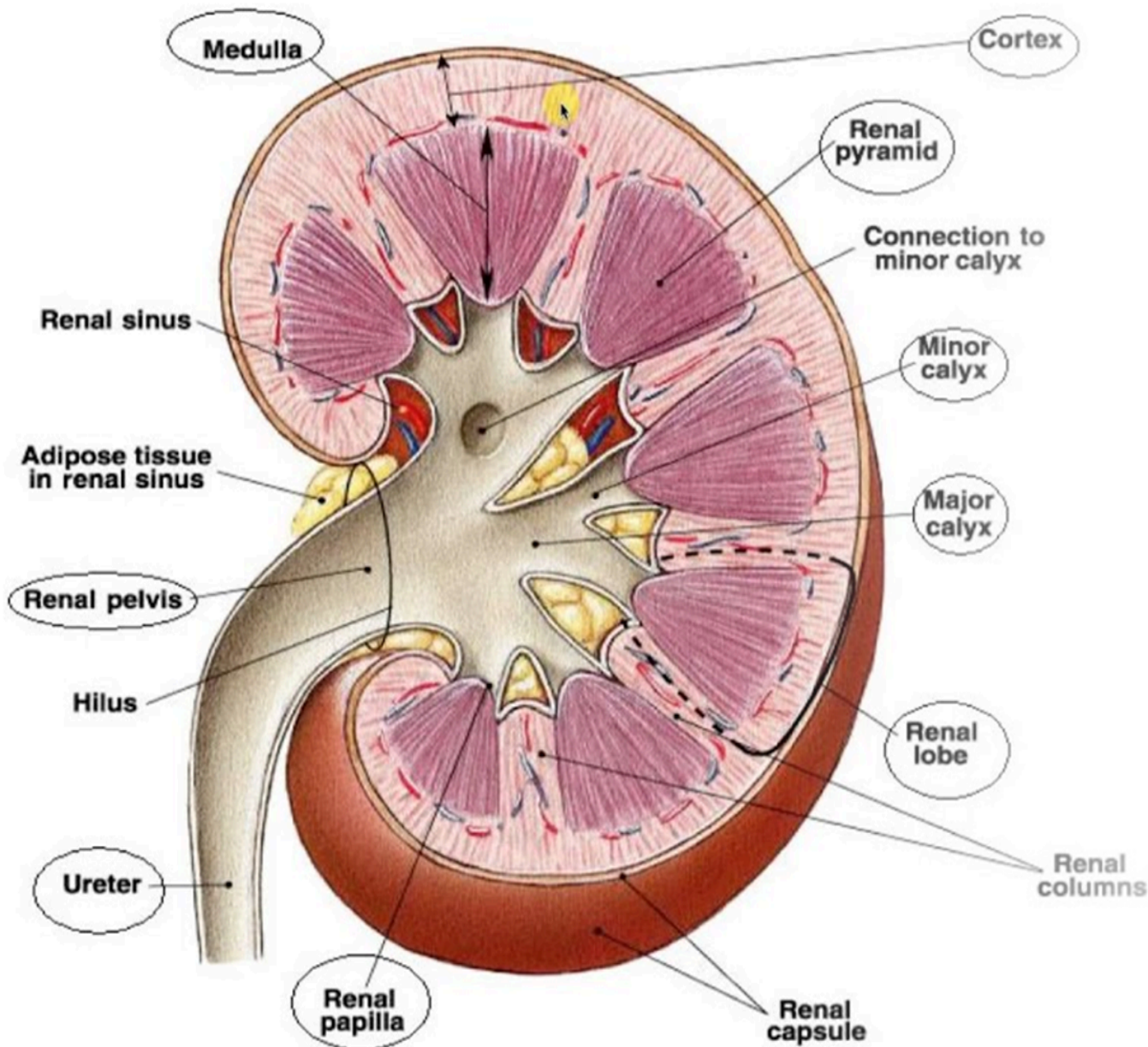
- Describe the histological structure of The glomerular filtration barrier.

• Histological Structure Of the Kidney

- *Examination with the naked eye* of a fresh, hemisected kidney reveals that its substance can be divided into two distinct regions:
- **Cortex**, the outer reddish brown part
- **Medulla**, the much lighter colored inner part.







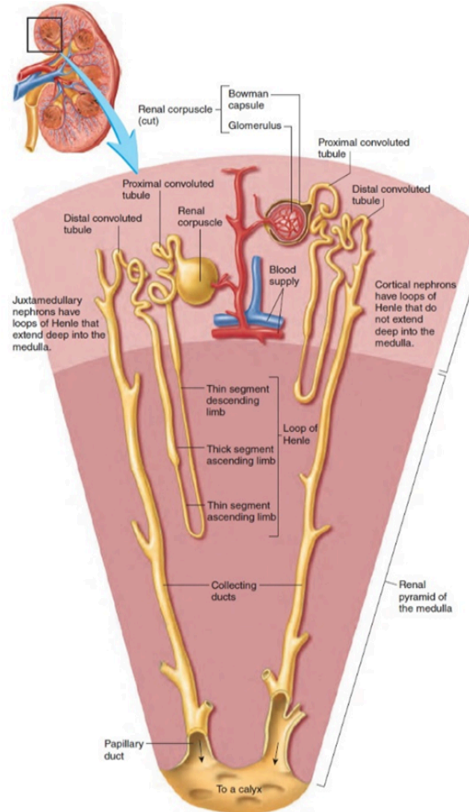
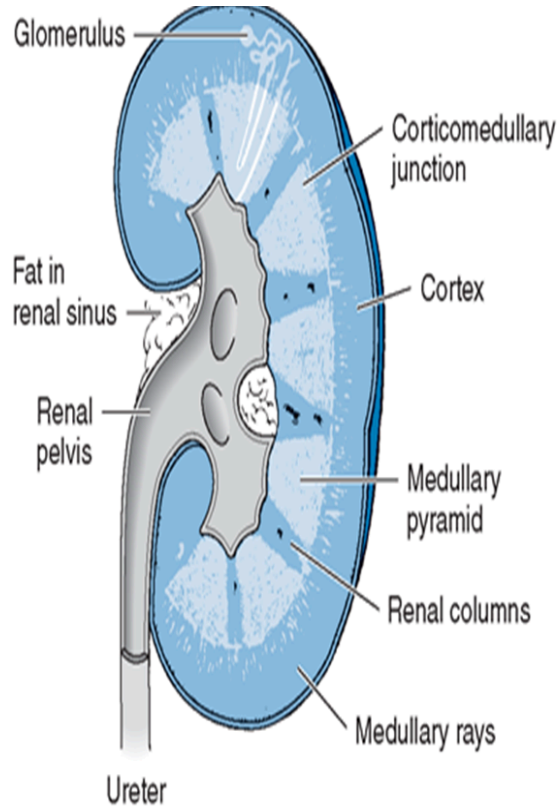
• **GENERAL STRUCTURE OF THE KIDNEY:**

- Each kidney is bean-shaped, with a concave hilum where the ureter and the renal artery and veins enter.
- The ureter divides and subdivides into several major and minor calyces, around which is located the renal sinus containing adipose tissue.

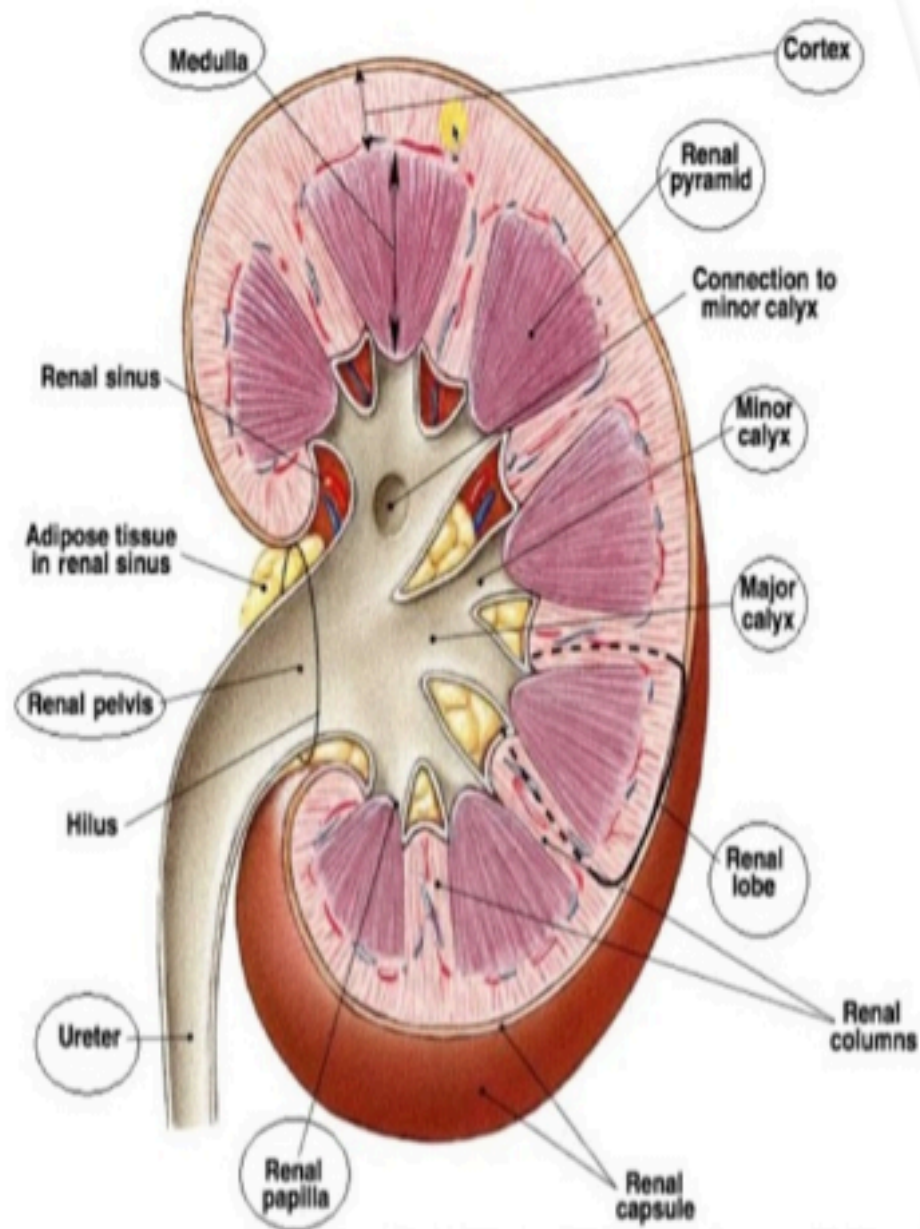
- Each kidney is surrounded by a thin *fibrous connective tissue capsule* weakly attached to the surface of the kidney. The capsule consists of two layers:
 - an outer layer of fibroblasts and collagen fibers,
 - and an inner layer with a cellular component of myofibroblasts that becomes continuous with the connective tissue forming the walls of the calyces and renal pelvis .
- *The interstitial connective tissue* is scanty.



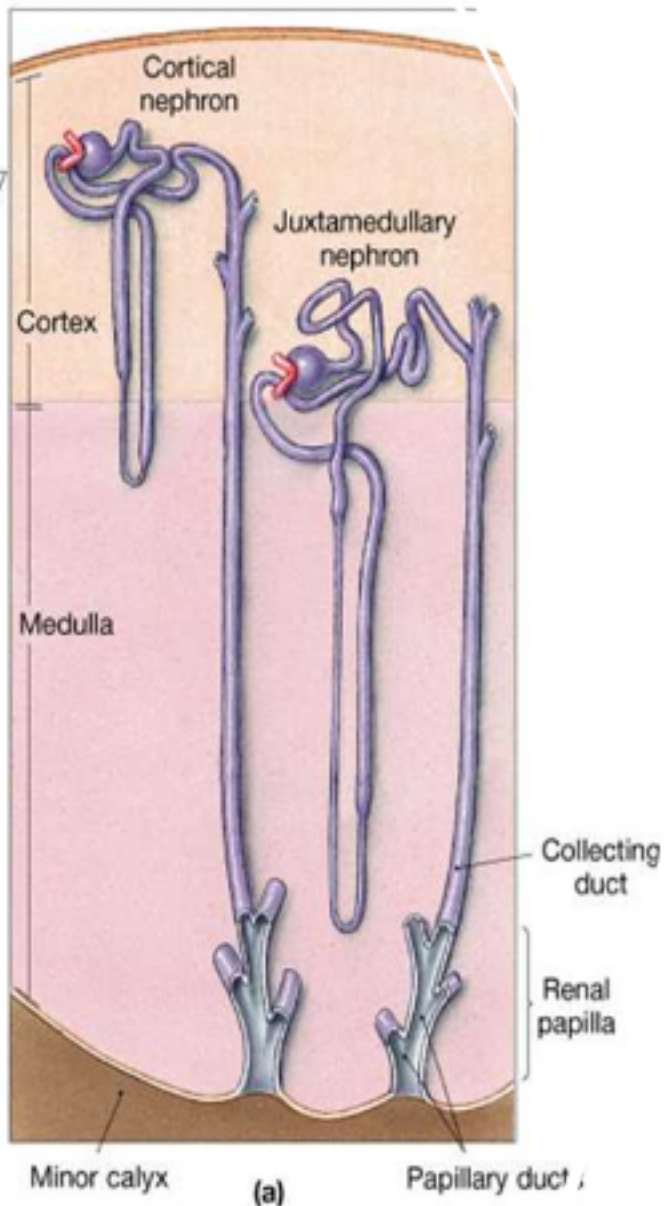
- *The parenchyma*
- -Formed of *an outer renal cortex* with many round corpuscles and tubules cross sections, and *an inner renal medulla* consisting mostly of aligned linear tubules and ducts.



- *The renal medulla* in humans consists of 8 –15 conical structures called ***renal pyramids***, all with their bases meeting the cortex (at the corticomedullary junction) and separated from each other by extensions of the cortex called ***renal columns***.



- -lobe. Each pyramid plus the cortical tissue at its base and extending along its sides constitutes a **renal lobe**
- -Parallel ducts and tubules extending from the medulla into the cortex comprise the **medullary rays**; these plus their associated cortical tissue are considered **renal lobules**.
 - The tip of each pyramid, called the **renal papilla**, projects into a minor calyx that collects urine formed by tubules in one renal

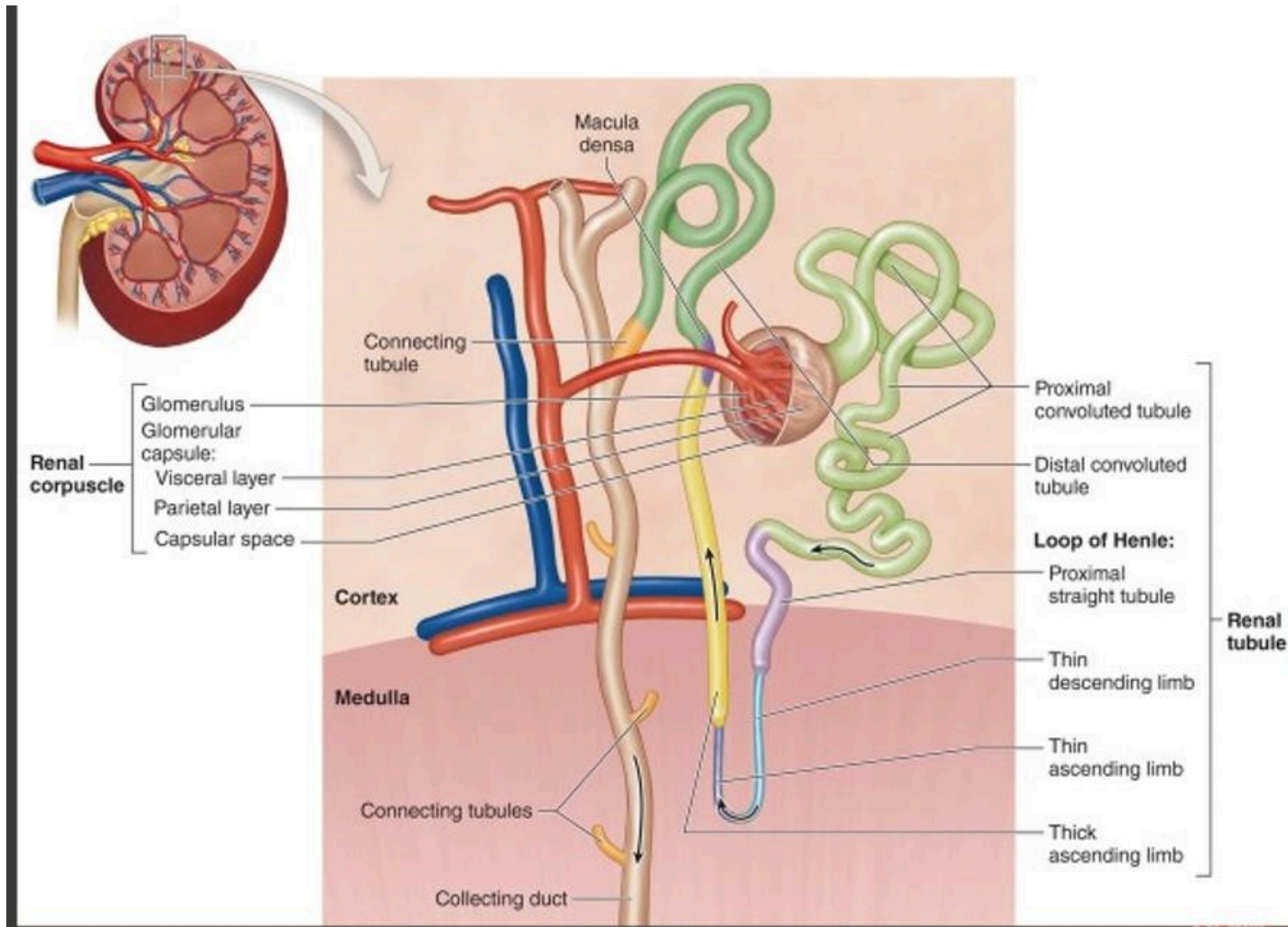


- Each Kidney contains 1–1.3 million functional units called ***nephrons***.
- -Each nephron consists of ***a renal corpuscle*** and ***a long, simple epithelial renal tubule*** with three main parts along its length. The nephron is a tube about 55 mm in length in the human kidney.
- -The number of nephrons decreases slightly in older adults, a process accelerated by high blood pressure.
- **Types :**
- -***Cortical nephrons*** are located almost completely in the cortex
- while ***juxtamedullary nephrons*** (about one-seventh of the total nephrons) lie close to the medulla and have long loops of Henle.
- -

If a kidney is donated for transplant (unilateral nephrectomy), the remaining kidney undergoes compensatory growth with cellular hypertrophy in the proximal parts of the nephron tubules and an increase in the rate of filtration, which allow normal renal function .

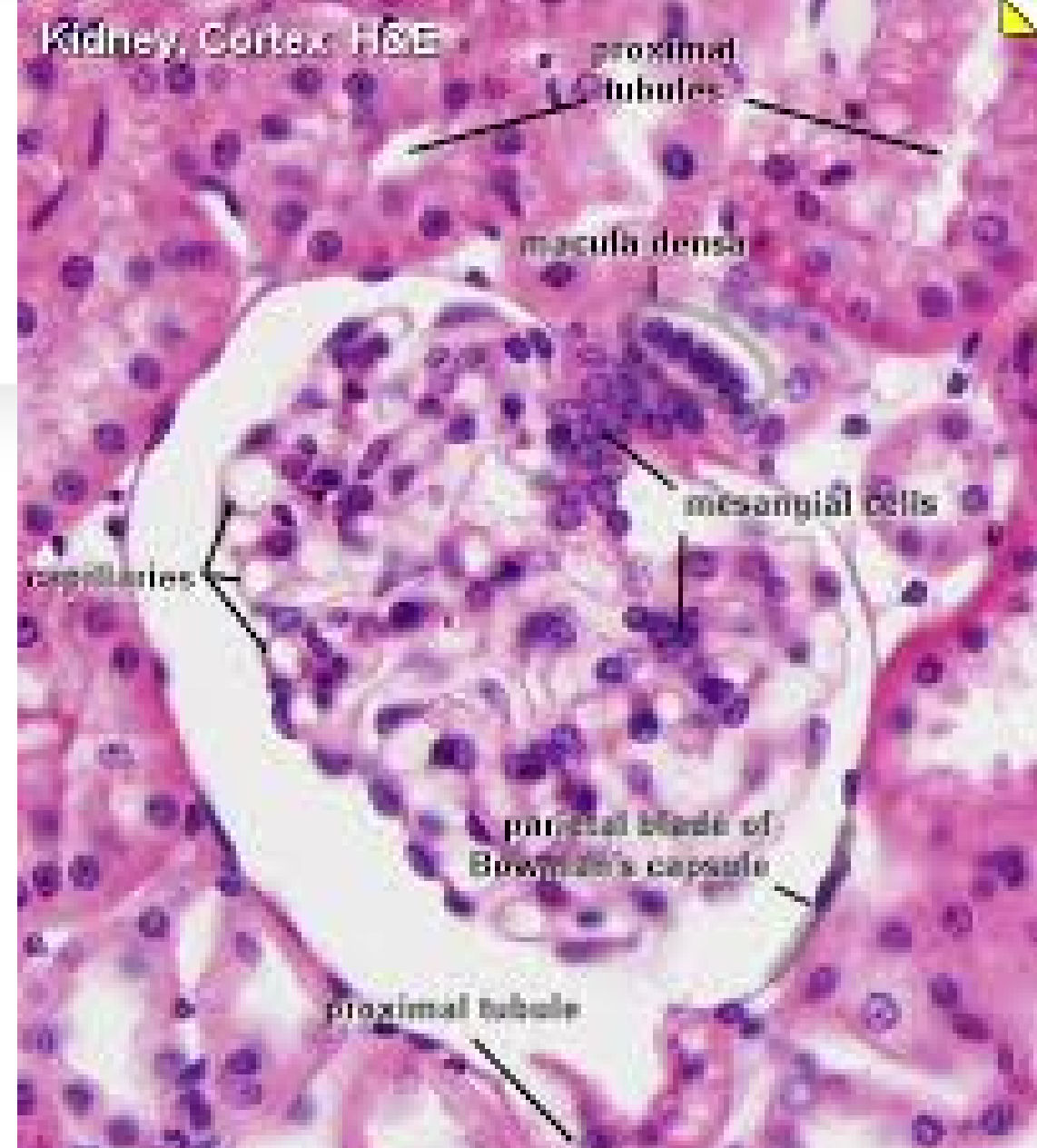
Nephron

- *The major divisions of each nephron are:*
 - **Renal corpuscle**;
 - **Proximal tubule**, a long convoluted part, located entirely in the cortex,
 - with a shorter straight part that enters the medulla;
 - **Loop of Henle** (or nephron loop), in the medulla, with a thin descending and a thin ascending limb;
 - **Distal tubule**, consisting of a thick straight part ascending from the loop Of Henle back into the cortex and a convoluted part completely in the cortex; and
 - **Connecting tubule**, a short minor part linking the nephron to collecting duct . Connecting tubules from several nephrons merge to form collecting tubules that then merge as larger **collecting ducts**. These converge in the renal papilla, where they deliver urine to a **minor calyx**.



Renal Corpuscles

- The renal corpuscle is the initial dilated part of the nephron enclosing a tuft of capillary loops.
- It is the site of blood filtration, always located in the cortex, about 200 μm in diameter.
- It consists of a tuft of glomerular capillaries, *the glomerulus*, surrounded by a double-walled epithelial capsule called the *glomerular (Bowman) capsule*.
- It has two poles; a vascular pole & a urinary one.



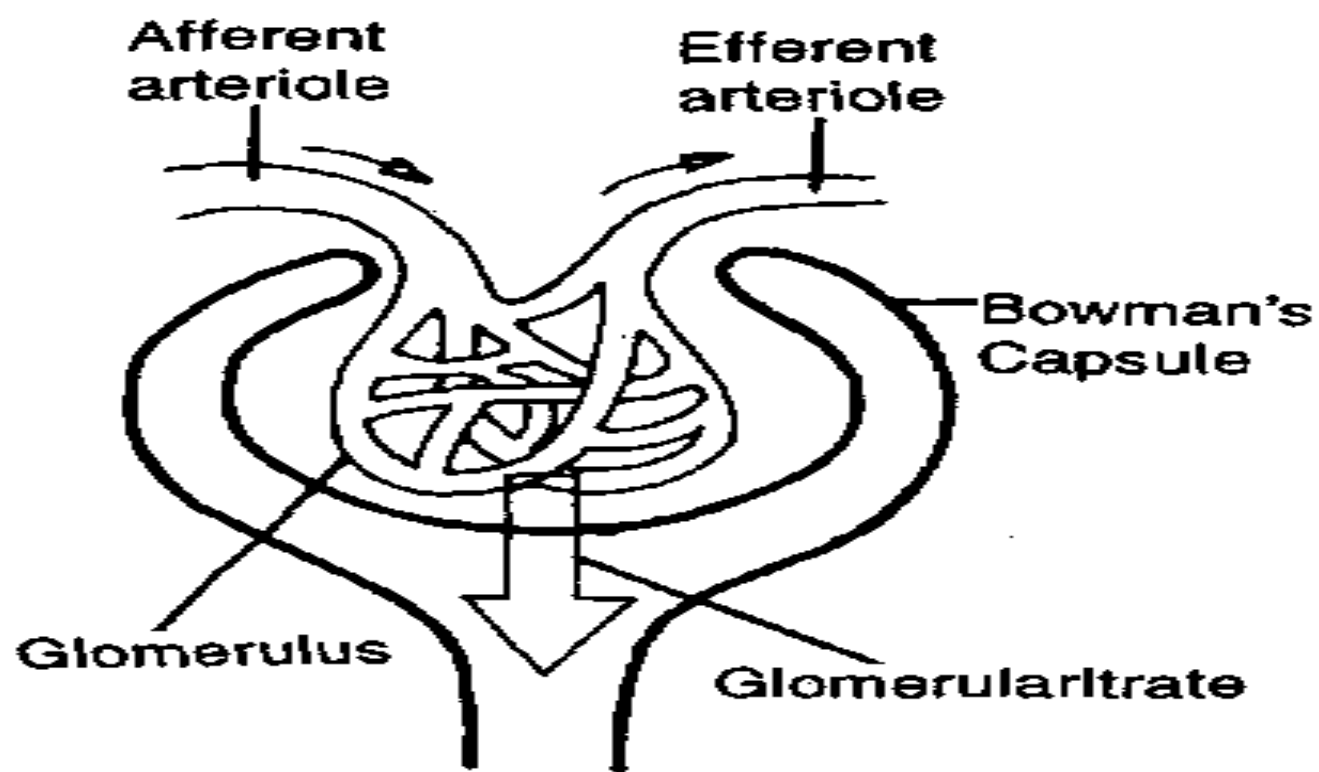
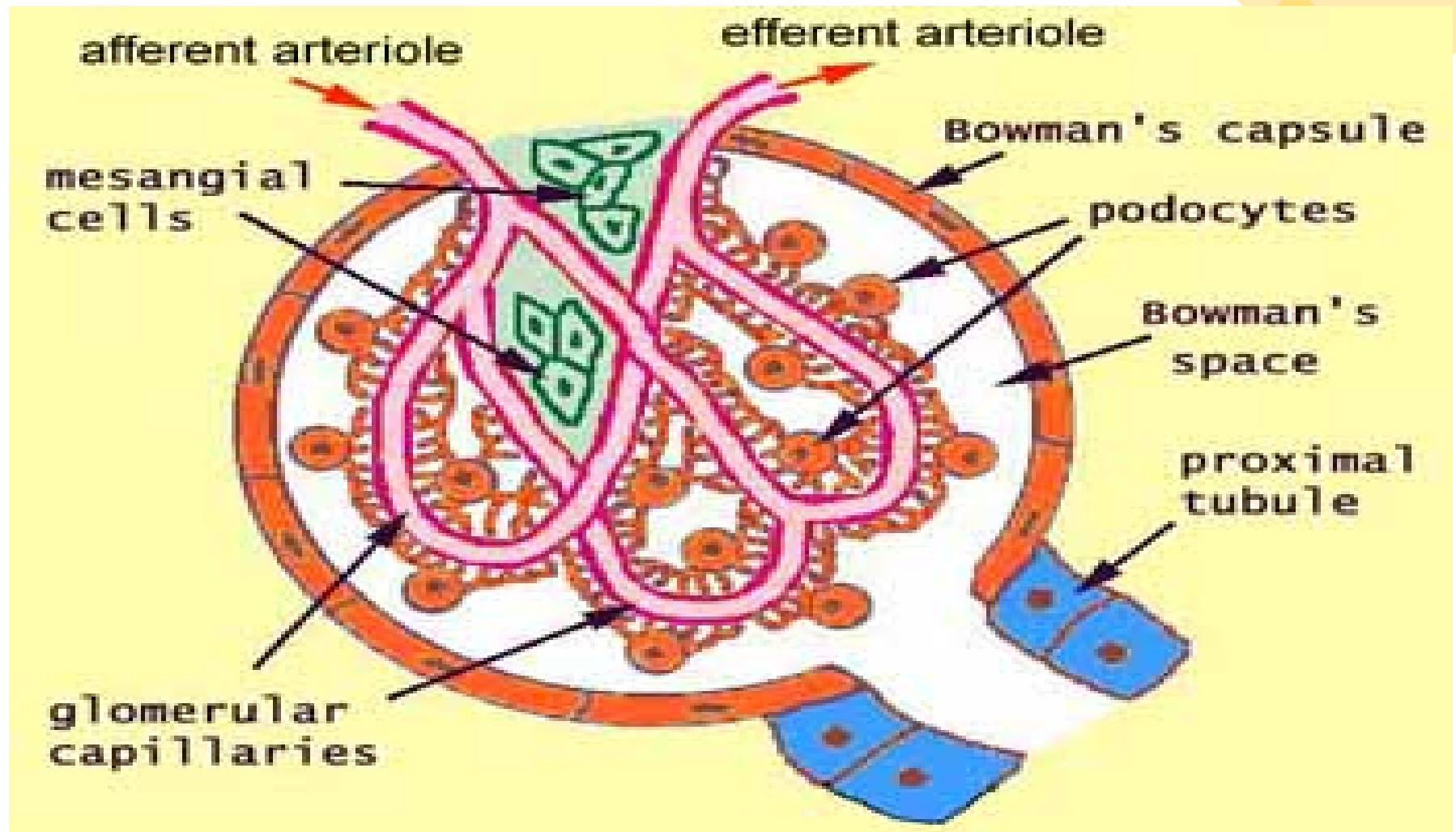
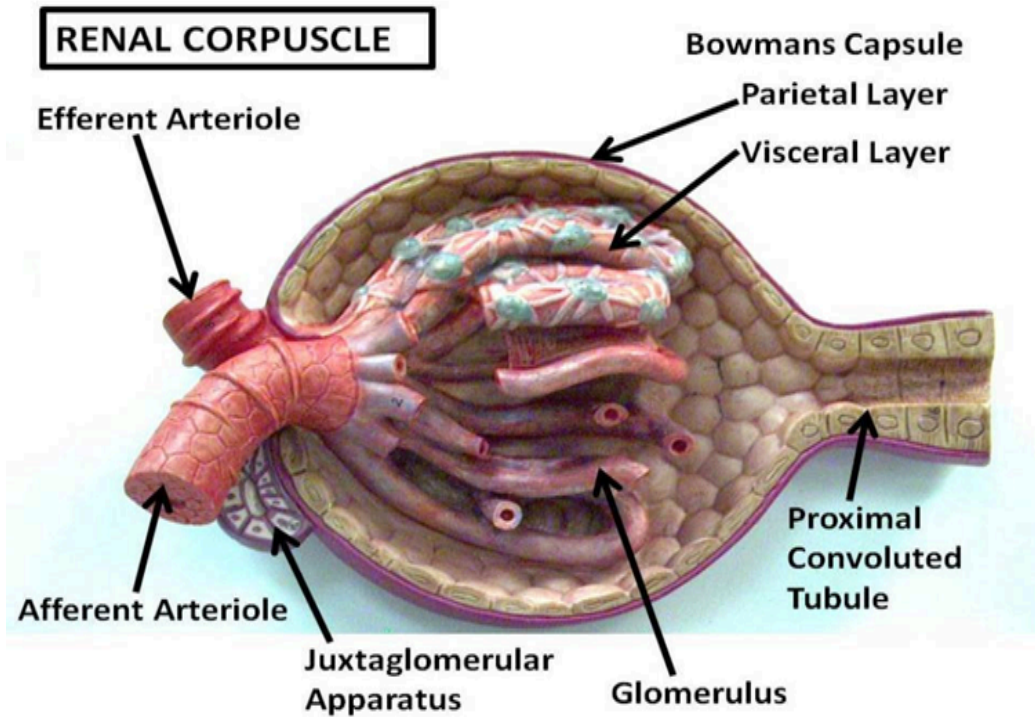


Fig. 4.6 Diagrammatic sketch of one malpighian capsule (Bowman's capsule + glomerulus)

A) Bowman's capsule:

- It is a double-walled cup, formed of a parietal and a visceral layers separated by Bowman's space or capsular space, which is continuous with the lumen of the renal tubule.
- ***The outer parietal layer of a glomerular capsule*** consists of a simple squamous epithelium supported externally by a basal lamina. At the tubular pole, this epithelium changes to the simple cuboidal epithelium that continues and forms the proximal tubule.
- ***The visceral layer of the glomerular (Bowman) capsule*** consists of unusual stellate epithelial cells called ***podocytes***, From the cell body of each podocyte several primary processes extend and curve around a length of glomerular capillary.

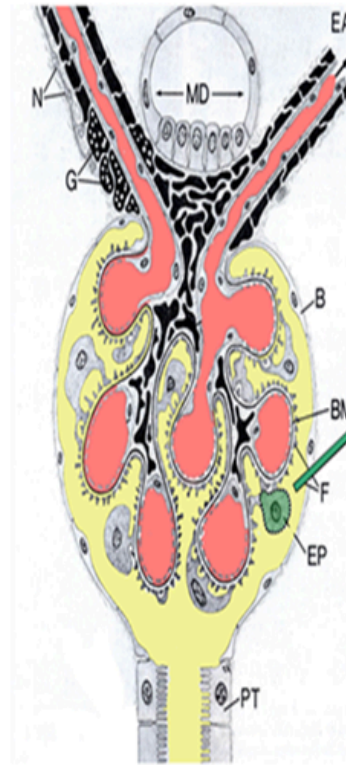


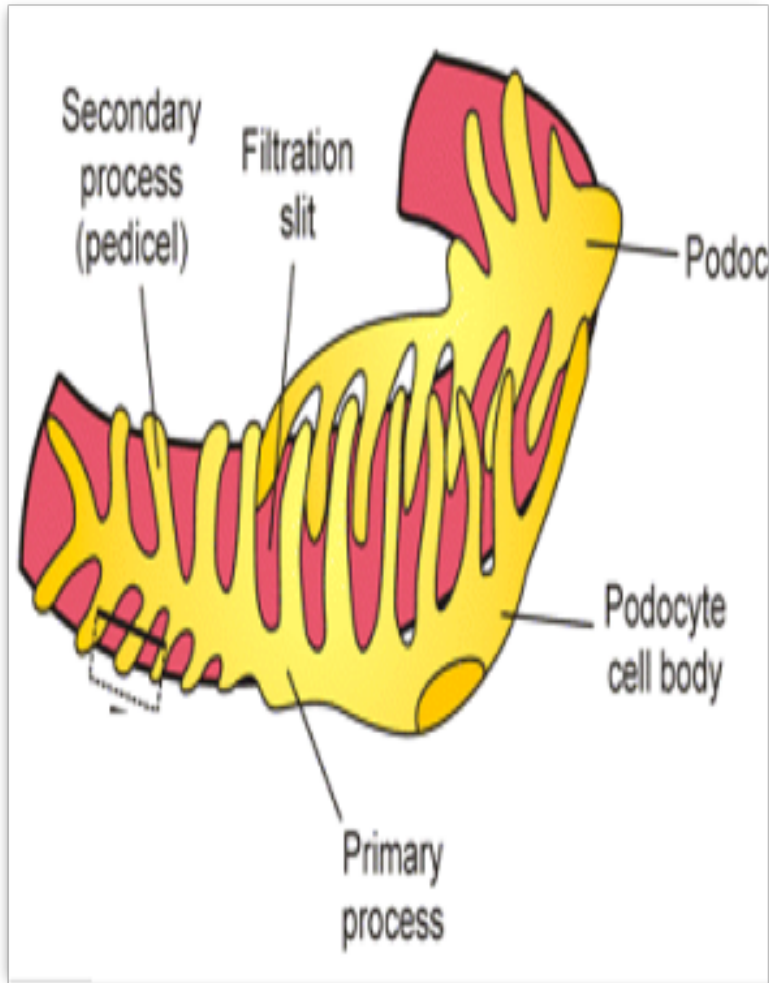


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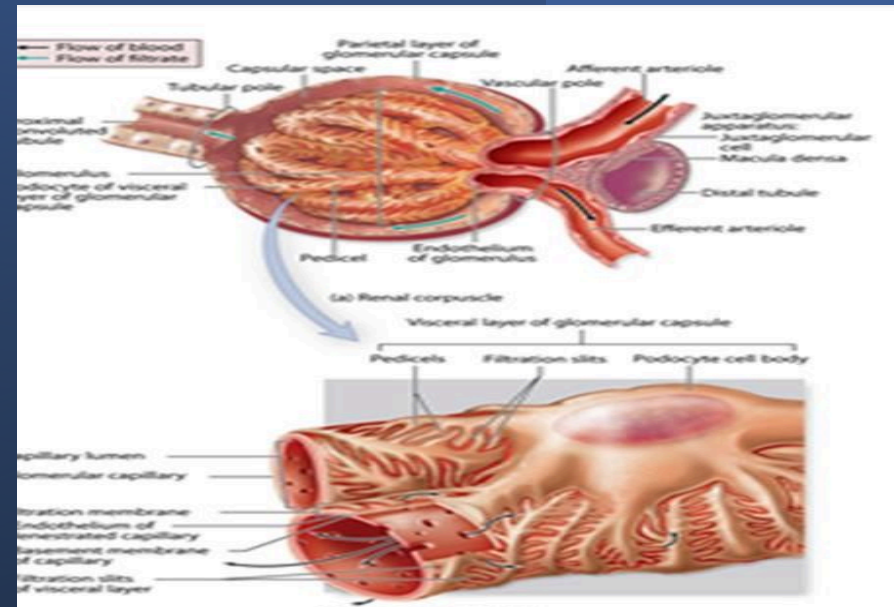
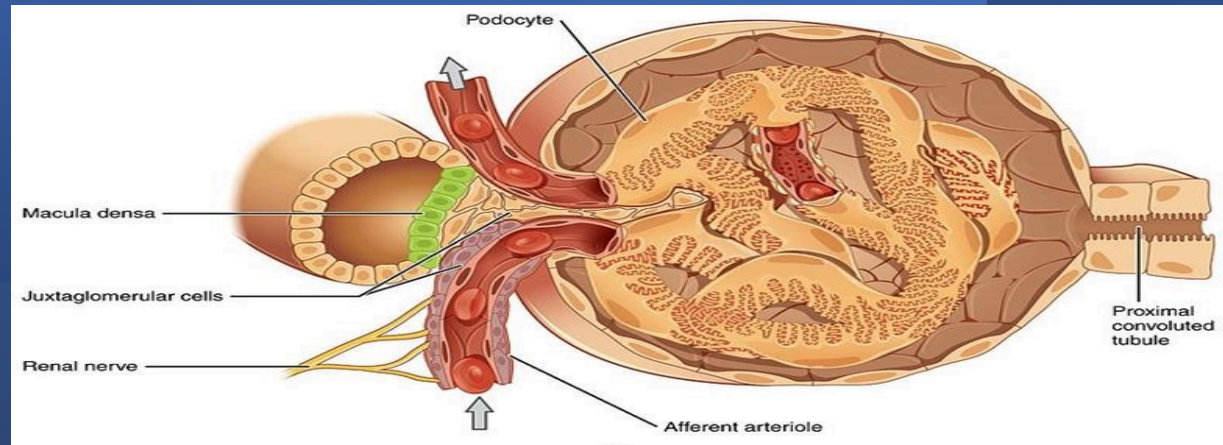
podocytes

- Each primary process gives rise to many parallel, interdigitating **secondary processes or pedicels** (L. pediculus, little foot).
- They interdigitate with the secondary processes of adjacent podocytes forming the **filtration slits**, bridged by **slit diaphragms** (formed of the glycocalyx of the pedicels).
- Podocyte foot processes are contractile (they contain actin and myosin).
- They are connected to each other by the slit diaphragm and to the basal lamina.
- The slit diaphragm molecular complex is associated with the actin cytoskeleton. Alterations in composition and/or arrangement of these complexes are found in many forms of human and experimental diseases.





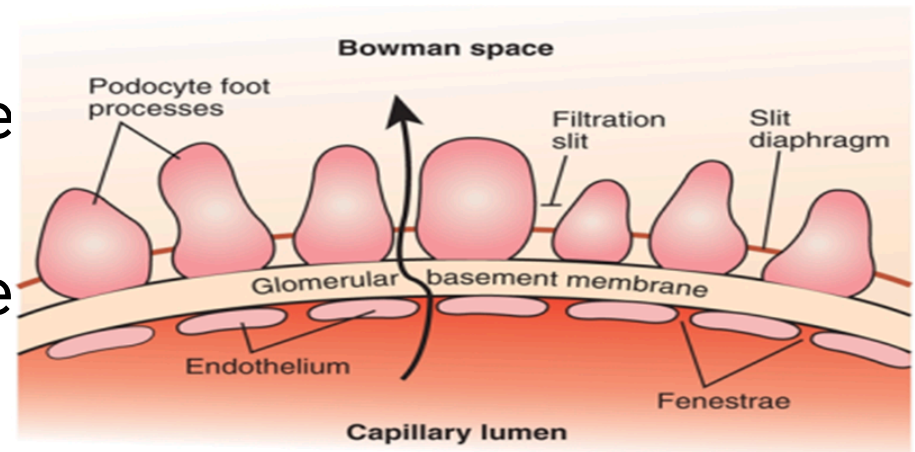
- The pedicels cover much of the capillary surface, in direct contact with the basal lamina. Between the interdigitating pedicels are elongated spaces, or filtration slit pores, 25- to 30-nm wide.
- -Spanning adjacent pedicels and bridging the slit pores are zipper-like slit diaphragms.



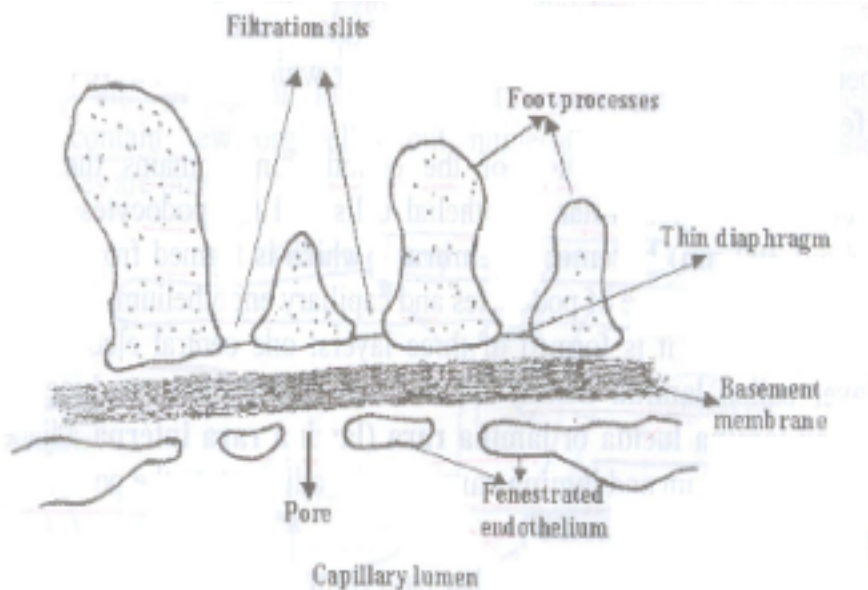
The renal corpuscle

- **B) Glomerular capillaries:**

- -Have **highly fenestrated endothelial** cells with a total length of approximately 1 cm and are uniquely situated between two arterioles (afferent and efferent) the muscle of which allows increased hydrostatic pressure in these vessels, favoring movement of plasma across the glomerular filter.
- -Between the endothelial cells of the capillaries and the covering podocytes is a thick (300–360 nm) **glomerular basement membrane (GBM)**. This membrane is the most substantial part of the filtration barrier (that separates the blood in the capillary lumen from the capsular space) and is formed by fusion of the capillary- and podocyte- basal laminae.
- .- Laminin and fibronectin in this fused basement membrane bind integrins of both the podocyte and endothelial cell membranes with the meshwork of type IV collagen

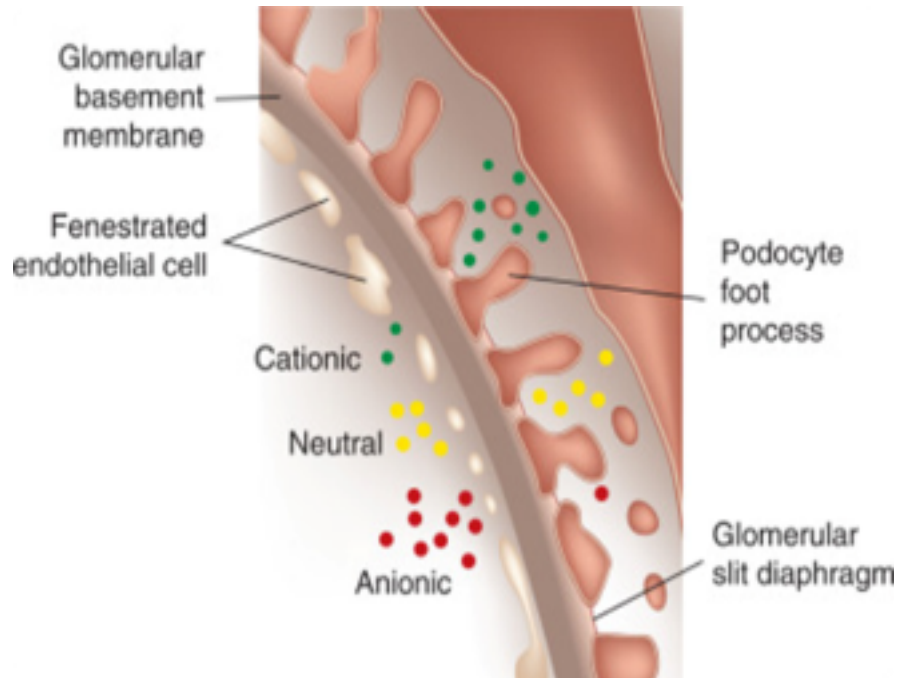


The glomerular filtration barrier

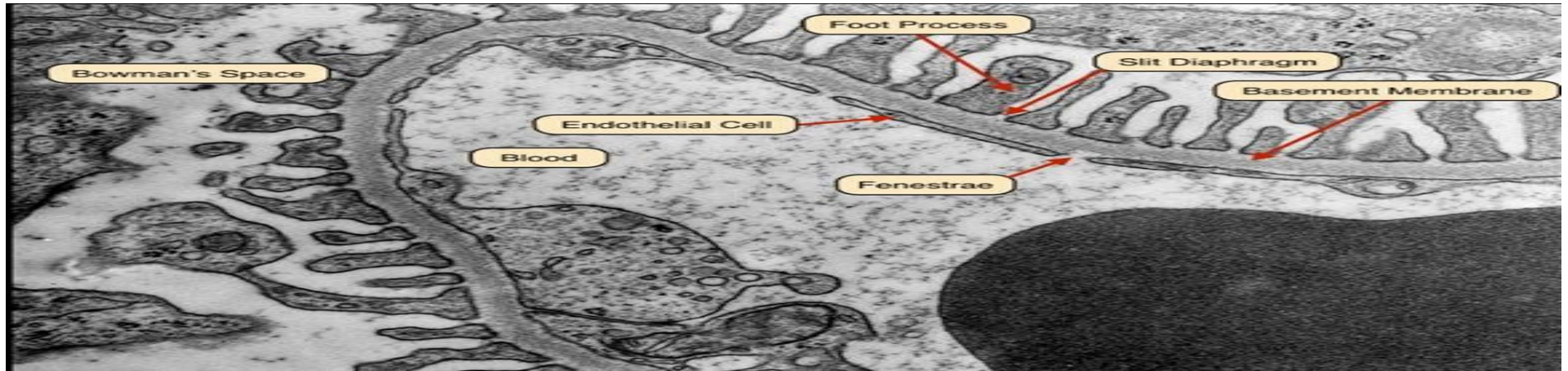


. Diagram of blood renal barrier.

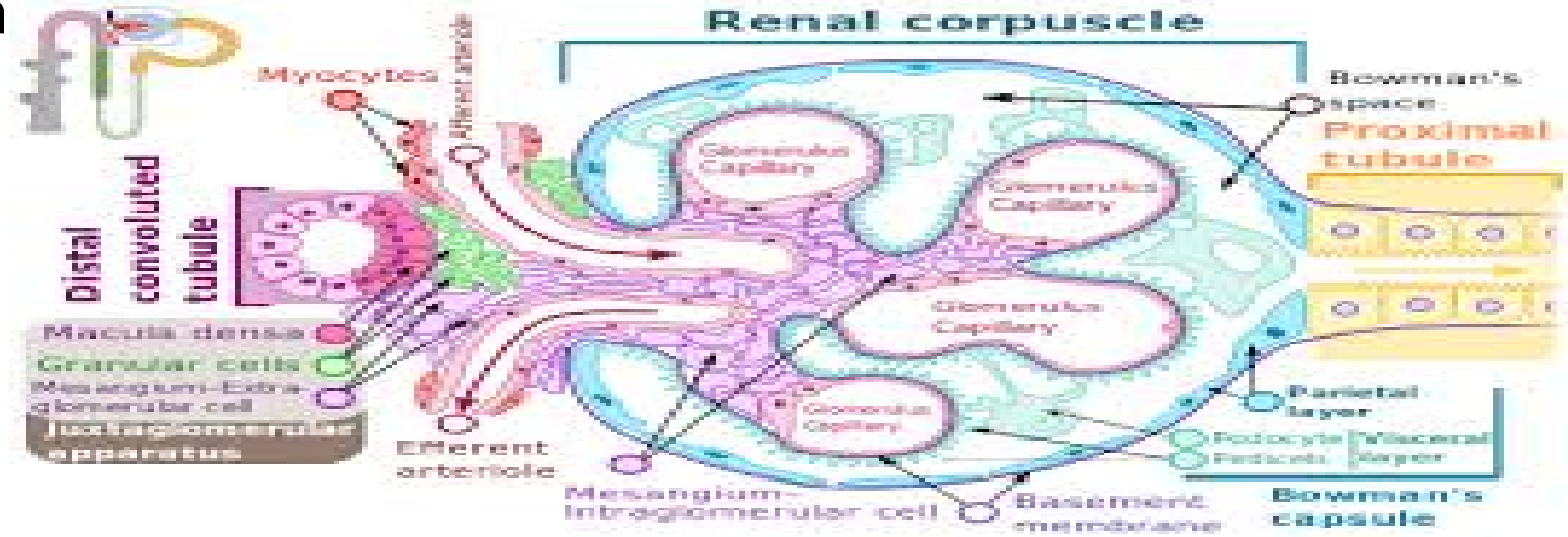
- The glomerular filtration barrier consists of three layered components:
- 1-The fenestrated capillary endothelium,
- 2-The glomerular basement membrane (GBM), the major component of the filter formed by fusion of the basal laminae of a podocyte and a capillary endothelial cell
- 3-Filtration slit diaphragms between podocytic pedicels.



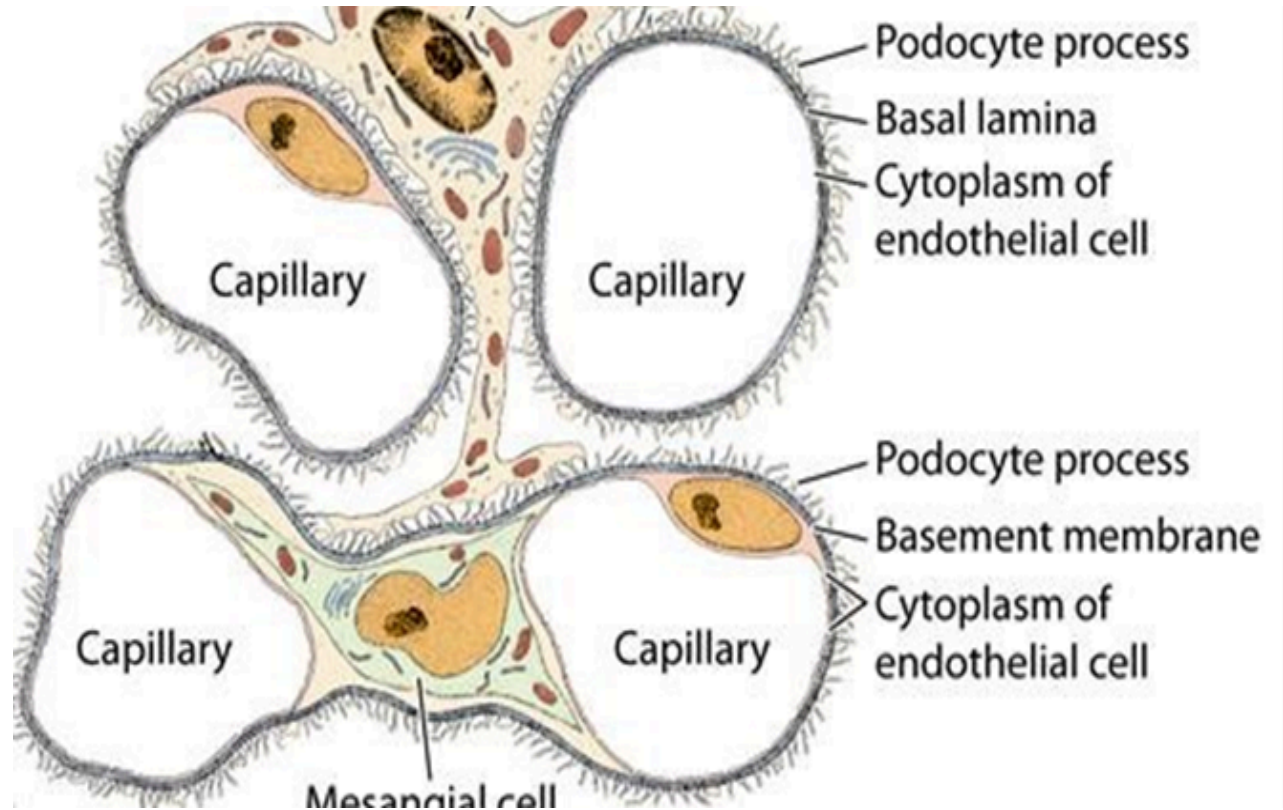
- Fenestrations block the exit of cells, but allow free flow of plasma.
- The shared basal lamina of podocytes and endothelium constitutes the first, coarser filtration barrier; it blocks the passage of molecules larger than 70 kD.
- The thin diaphragms covering the slit openings between the podocyte foot processes constitute a more selective filter.



- In addition to capillary endothelial cells and podocytes, renal corpuscles also contain *mesangial cells* (Gr. Mesos, in the midst + angion, vessel), most of which resemble vascular pericytes in having contractile properties and producing components of an external lamina. Mesangial cells are difficult to distinguish in routine sections from podocytes, but often stain more darkly.
- -They and their surrounding matrix comprise the mesangium, which fills interstices between capillaries that

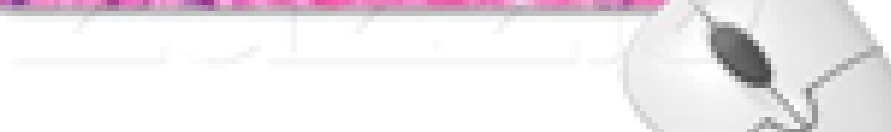


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- Mesangial cells extend contractile processes along capillaries that help regulate blood flow in the glomerulus. Some mesangial processes pass between endothelial cells into the capillary lumen where they may help remove or endocytose adherent protein aggregates



- *Functions of the mesangium include the following:*

- . 1-Physical support of capillaries within the glomerulus.
- . 2-Adjusted contractions in response to blood pressure changes, which help
- 3-maintain an optimal filtration rate.
- . 4-Phagocytosis of protein aggregates adhering to the glomerular filter, including antibody-antigen complexes abundant in many pathological conditions
- . .5-Secretion of several cytokines, prostaglandins, and other factors important for immune defense and repair in the glomerulus





References:



Junqueira's Basic Histology (Text and Atlas) (2018) 15 th edition.



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QUIZ

Choose the correct answer

1. Regarding the cortical nephrons which is the distinguishing character?
 - a) They constitute about 85% of all nephrons
 - b) They have long loops of Henle
 - c) They have vasa recta
 - d) They don't have peritubular capillaries